

CS 301 Midterm — S002 — Question 1

QuickSort divides the array using a pivot.

Average is $O(n \log n)$ but the worst case $O(n^2)$ occurs when pivots are poorly chosen.

The partition step is linear $O(n)$.

CS 301 Midterm — S002 — Question 2

BFS goes broad first, uses a queue.

Good for finding shortest path.

DFS goes deep first, uses stack.

Useful for cycle detection.

Both run in $O(V + E)$ time.